

Generative Artificial Intelligence (GenAI)

Capabilities and Limitations of New GenAI Tools



CCST
CALIFORNIA COUNCIL ON
SCIENCE & TECHNOLOGY

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Backgrounder | April, 2025

What is GenAI?

Generative Artificial Intelligence (GenAI) tools such as ChatGPT, Claude, Gemini, and DALL-E create new text, image, audio, and video content from prompts inputted by users. GenAI uses new advances in deep-learning models to analyze patterns in the training data and then generate new work (e.g., a picture of a dog wearing a hat) that is informed by the training data, but that can be very different from what it has been trained on.

This ability to create wholly new content from an input prompt (e.g., create a picture of a dog wearing a hat) is what distinguishes GenAI from other AI use cases. For example, AI is also commonly used to accomplish classification tasks of content that already exists (e.g., find all pictures of a dog wearing a hat in a photo album).

Despite the many potential benefits of automating, accelerating, and augmenting tasks traditionally performed by humans, GenAI tools are associated with many significant risks stemming both from how they are developed and how they are deployed. Furthermore, GenAI tools and the infrastructure networks that support them can consume considerable amounts of energy and water to run the massive amounts of needed computations.

How Does GenAI work?

To develop GenAI tools, computer programmers use so-called foundation models, or machine learning models, that are trained on a broad set of data for different tasks which can then be fine-tuned to serve as the foundation for many applications. As a type of foundation model, large language models (LLMs) are specifically designed to work with language. They are classified as “large” based on the number of parameters used in the model. Parameters are the factors that a model uses to generate its outputs, and are adjusted and refined as the model is trained on a data set. For example, Open AI’s GPT-4 LLM foundation model is made up of nearly 2 trillion parameters.

The foundation models that GenAI applications are built upon represent a significant step in the evolution of previous AI models because they can process extremely large and varied sets of unstructured data and can perform a wide range of tasks despite not being explicitly trained for the task. However, the immense amounts of computing power and data required, especially during model training, has meant that the majority of GenAI tools are being developed by a small number of private companies.

Although the capabilities of GenAI are astounding, it is important to appreciate that these tools are not creative in the same sense as human creativity. GenAI tools are, in a sense, extraordinarily capable mimics that are dependent on the quality of the training data and the penalty/reward system that was used to teach the model which outputs qualified as “correct” during model training. Understanding the capabilities and limitations of GenAI is important for understanding the potential use cases and risks of GenAI technologies in various applications.



GenAI example. Prompt: “A photo realistic image of a dog wearing a hat in front of the California State Capitol.”
Result: An output image of a very good dog in front of a building that somewhat resembles the State Capitol.
Created by CCST staff using Canva’s Magic Media.



Example GenAI Use Cases

Consumer Sales & Marketing

- **Personalized advertising and outreach:** Generate tailored advertisements, social media posts, and chatbot assistants to relevant audiences.
- **Marketing analytics:** Generate a range of possible page layouts or ad copy to test various marketing strategies.
- **Price negotiation:** Generate recommended offers during competitive pricing negotiations.

Banking & Finance

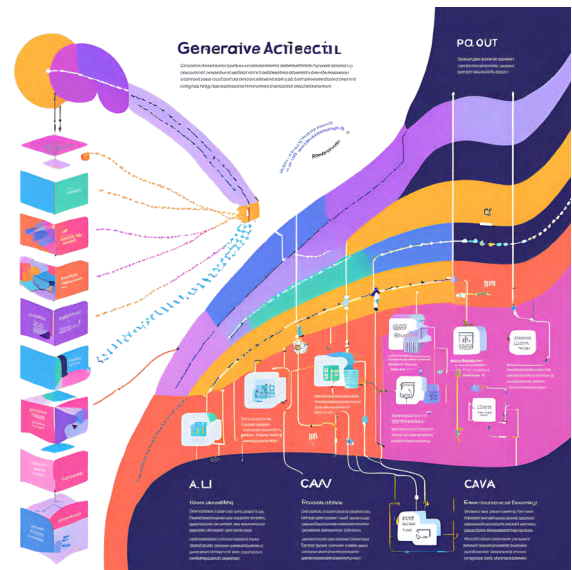
- **Fraud detection and risk assessments:** Generate synthetic data that mimic real-world scenarios to test fraud detection systems and models that assess customer credit risk.
- **Compliance:** Generate summaries of regulatory reports for banking compliance officers to act on.
- **Investment strategy:** Generate advice for clients on purchasing equities and or investment products.
- **Technical documents:** Generate written technical documents such as audit reports, environmental, social, and governance (ESG) reports, or loan and mortgage contracts.

Medical & Healthcare

- **Drug discovery:** Generate novel molecules based on desired properties or generate synthetic patient data to predict the outcomes of clinical trials testing new drug candidates.
- **Medical and surgical simulation:** Generate virtual patient simulations to allow medical students and professionals to practice procedures and diagnostic environments (e.g., detecting cancer or other abnormalities in MRI or CT scans).
- **Personalized or precision medicine:** Generate personalized treatment plans (e.g., medicine combinations, surgical options etc.) based on patient intake data and medical history.

Education

- **Course content creation:** Generate personalized learning plans, sample problems, writing prompts, and tests based on a student's performance and progress.
- **Course evaluations:** Generate recommended grades for assignments or tests and generate detailed feedback for students.
- **Language translation:** Generate translations of educational materials into different languages to assist learners in their native language.



GenAI example. Prompt: "A diagram illustrating the details of how generative AI technology works" Result: A non-sensical diagram with no accurate information. Created by CCST staff using Canva's Magic Media.

Catch Up on CCST AI Briefings

- **AI Tools for Addressing Conservation and Biodiversity:**
AI for conservation efforts, through real-time monitoring, predictive modeling, and targeted species protection, including recent technological advancements, the challenges of implementing AI solutions, and their implications for policymakers.
March 4, 2025.
- **The Role of AI in Addressing our Natural Resources Challenges:**
Current and emerging applications of AI in the natural resources, with an eye toward data infrastructure and workforce adaptation.
Oct. 17, 2024.
- **Leveraging Artificial Intelligence for a Resilient California:**
Exploring the untapped potential of AI technology to mitigate the impact of disasters via disaster preparedness, response, and recovery.
Feb. 5, 2024



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Risks & Concerns

GenAI tools have many potential benefits in automating, accelerating, and augmenting human capabilities, but they also have many significant risks stemming both from how they are developed and deployed. GenAI creations are outputs from complex models and come with no inherent guarantee that they are unbiased, correct, or accurately reflect reality.

Here we focus on some risks specific to GenAI technology. These examples are not exhaustive and do not consider “second order” impacts of this new technology on jobs or the economy at large.

Biased Outputs

GenAI models are only as good as the data they were trained on. Any biases, omissions, inconsistencies, or factually incorrect information contained within the training data will be propagated in the content created by GenAI. As a consequence, GenAI content can perpetuate inaccurate information or harmful racial, gender, and other stereotypes. Many GenAI tools are trained on data scraped from the internet and lack transparency on data sources or quality control checks.

Hallucinations

GenAI tools can output words or phrases that are nonsensical, grammatically incorrect, or factually false (despite sounding factually correct). Hallucinations can lead to the problematic spread of misinformation in the absence of fact-checking or quality control.

Fake Content

Advances in GenAI have made it easier to create high quality audio and visual mimics of real people, places, and things. As a consequence, it is becoming harder to distinguish between real content and so-called “deep-fakes” that could be misused by bad actors to propagate harmful misinformation.

Data Privacy

Training data for many GenAI tools often also includes sensitive data and personal information (e.g., names, addresses, health data). In many cases, individuals are unaware or have not consented to the use of their data in AI model training. When personal data is used in the training process, it is possible for it to be revealed in content outputted by the GenAI tool. In some cases, users successfully prompted GenAI tools to output prompts or inputs from other users. This poses particularly serious concerns for sensitive information such as health data when users interact with an AI medical chat bot.

Copyright & Intellectual Property

The eligibility of copyright protection for GenAI training data or outputs is a topic of ongoing debate. Much of the data used to train GenAI tools was scraped from publicly available internet sources of copyrighted material. Depending on the situation, it has been suggested that the programmer of the AI tool, the creator of the training data, or the user of the GenAI tool to create the outputs could all potentially be authors. In particular, debate has focused on what constitutes “fair use” when GenAI tools were trained using copyrighted material or used to create content that mimics the work of real authors and artists or to create a digital likeness of real people.

About CCST

The California Council on Science and Technology is a nonpartisan, nonprofit organization established via the California State Legislature in 1988. CCST engages leading experts in science and technology to advise state policymakers — ensuring that California policy is strengthened and informed by scientific knowledge, research, and innovation.

AI Policy Initiative

CCST’s AI Policy Initiative provides nonpartisan, science-informed insights to guide California’s policymakers through the rapidly evolving AI landscape. From navigating nuanced research recommendations to addressing societal impacts and ensuring equitable access, this initiative is poised to be a critical and timely resource for our state’s policymakers as they step up to the challenge of nation-leading action. CCST launched its Irwin Jacobs & Family AI Policy Funding Campaign, thanks to a seed donation from the Jacobs Family Fund of the Jewish Community Foundation.

Disaster Resilience Initiative

CCST’s Disaster Resilience Initiative is generously supported by the State of California through a one-time funding allocation in the 2021-2022 State budget. This allocation supports the transmission of information between science and technology experts and policymakers focused on the overlapping, intersecting, and compounding nature of disasters in California.

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